Roll No. | | | | | | | |

EVEN END SEMESTER EXAMINATION JUNE - 2025

Name of the Course: B Tech. Semester: IV

Name of the Paper: Fluid Mechanics and Fluid Machines Paper Code: **TME 407**

Time: 3 Hours Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Q1	(20marks)	**CO1, CO2, CO6**
(a)	Discuss the following properties of fluid: (i) kinematic viscosity, (ii) specific weight, (iii) bulk modulus, (iv) dynamic viscosity	
(b)	Calculate the density and specific weight of one litre of diesel of specific gravity = 0.8	
(c)	The space between two square flat parallel plates is filled with oil. Each side of the plate is 600 mm. The thickness of the oil film is 25 mm. The upper plate, which moves at 4 m/s requires a force of 140 N to maintain the speed. Determine (i) The dynamic viscosity of the oil (ii) The kinematic viscosity of oil if the specific gravity of oil is 0.85	
Q2	(20 marks)	**CO3, CO2, CO6**
(a)	Comment on the following: (a) path line, (b) circulation (c) temporal acceleration, (d) velocity potential function with necessary equation.	
(b)	Explain manometer and manometric head. Discuss U tube manometer and differential manometer with neat sketch.	
(c)	A clean tube of diameter 3 mm is immersed in a liquid with a coefficient of surface tension = 0.6 N/m. The angle of the liquid with the glass can be assumed to be 125^0. The density of the liquid = 12000 kg/m^3. What would be the level of the liquid in the tube relative to the free surface of the liquid inside the tube?	
Q3	(20 marks)	**CO3, CO2, CO6**
(a)	Write a short note on meta centric height. What is the importance of metacentric height in the stability of floating body? What would happen for a static fully submerged body if its center of buoyancy lies below center of gravity, please comment?	
(b)	State laminar, transient and turbulent flow. For the case of pipe flow, what is/are the conditions for a flow to be turbulent? Define temporal and convective acceleration.	
(c)	Water flows in a circular pipe. At one section the diameter is 0.3 m, the static pressure is 260 kPa gauge, the velocity is 3 m/s and the elevation is 10 m above ground level. The elevation at a section downstream is 0 m and the pipe diameter is 0.15 m. Find the gauge pressure at the downstream section. Frictional effect may be neglected. Assume density of water to be 999 kg/m^3.	
Q4	(20 marks)	**CO4, CO5, CO6**
(a)	Explain head loss in pipes. What are the minor and major losses occur in pipes, Please expalin	
(b)	Calculate the discharge through a pipe of diameter 200 mm when the difference of pressure head between the two ends of a pipe 500 m apart is 4 m of water. Take the value of f=0.009.	
(c)	Discuss the working principle of Francis turbine with neat sketch. What are the differences between Pelton turbine and Francis turbine?	
Q5	(20 marks)	

(a)	Write a short note on the following: (a) Pitot tube, (b) Venturimeter, (c) Orifice meter, (d) Nozzle, (e) Diffuser	**Specify COs for Q1 by Considering Bloom's Taxonomy**
(b)	Write a short note on net positive suction head (NPSH). What is the condition for NPSH to exist, please comment?	
(c)	Explain the working principle of axial pump with neat sketch.	